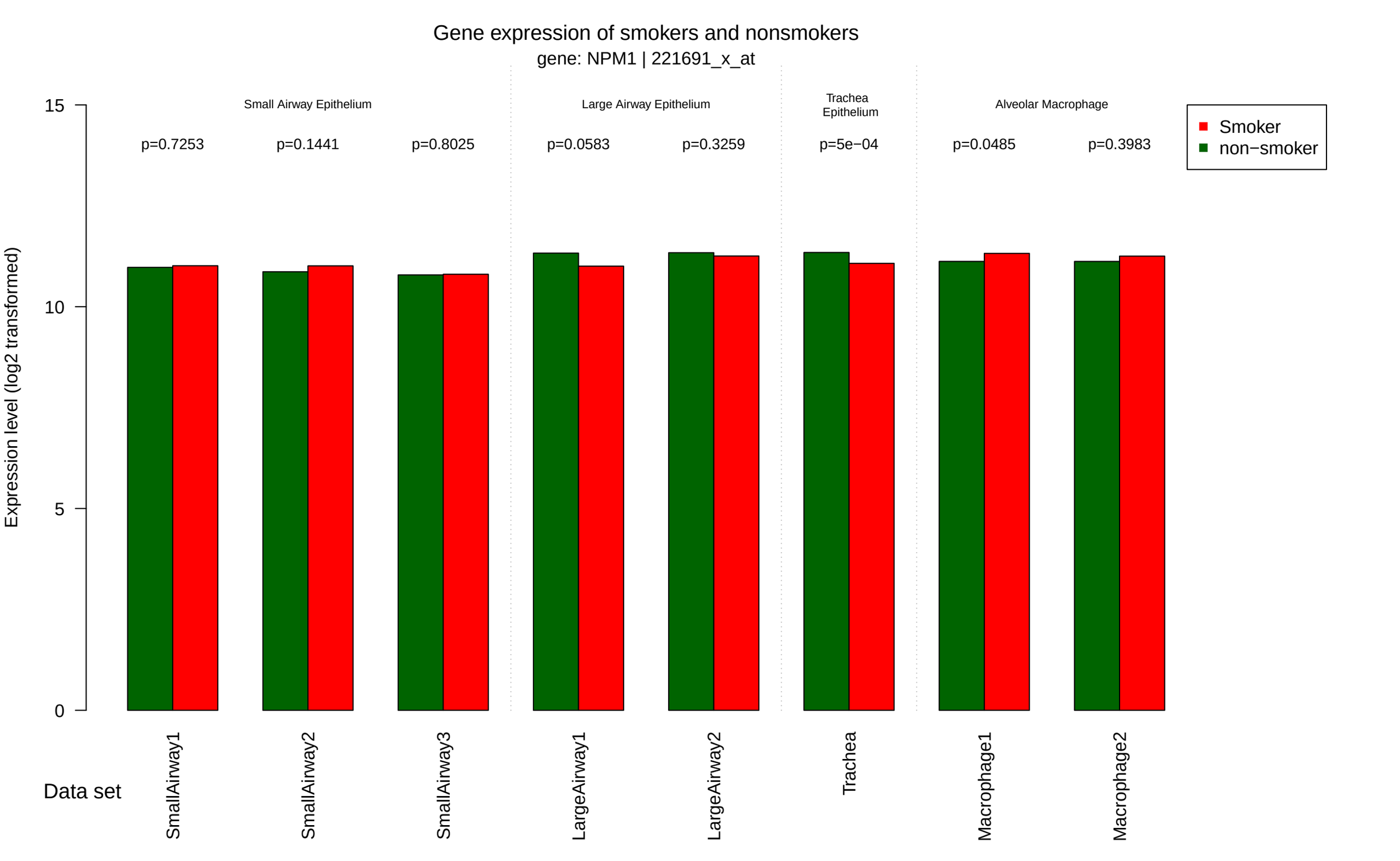
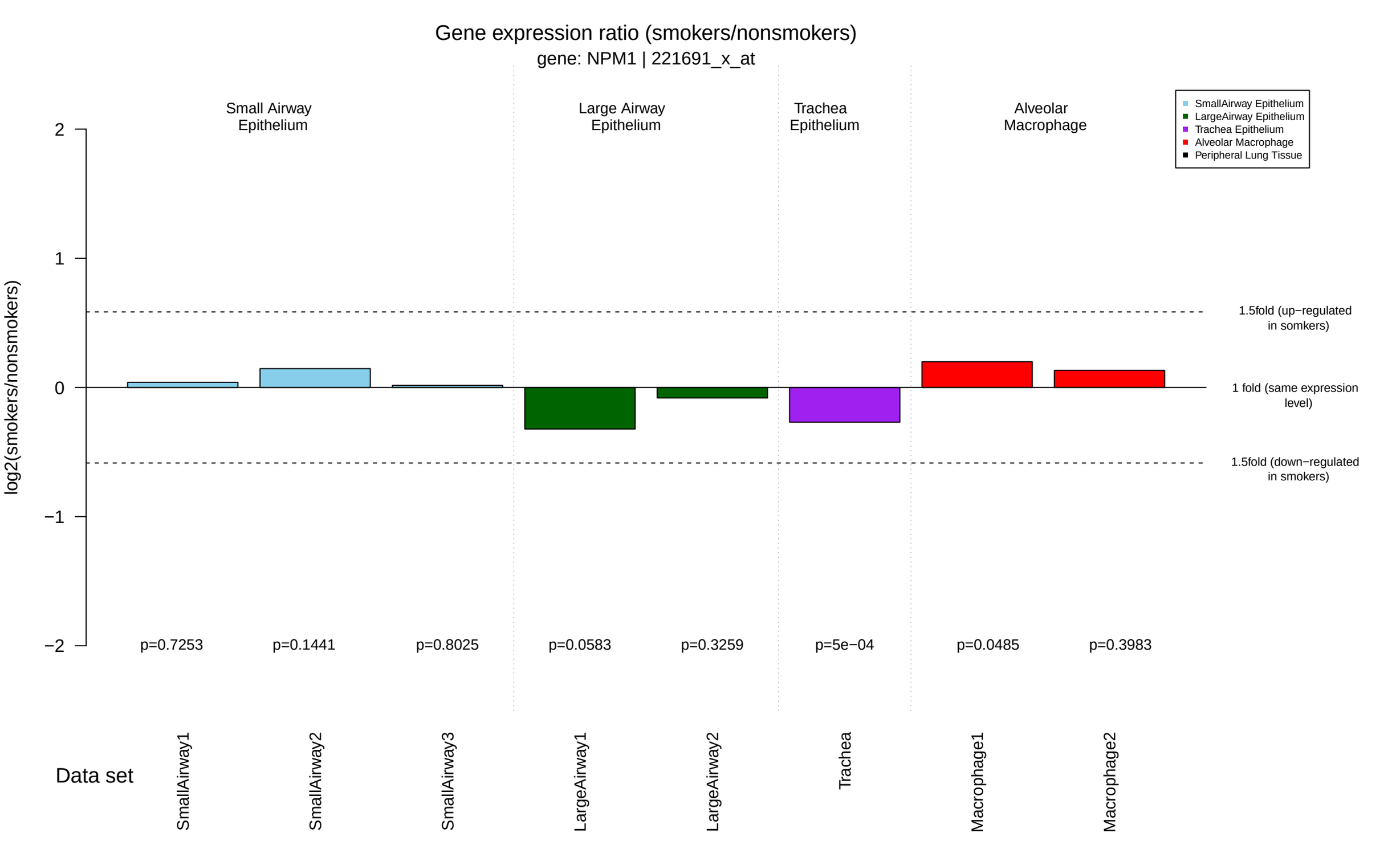
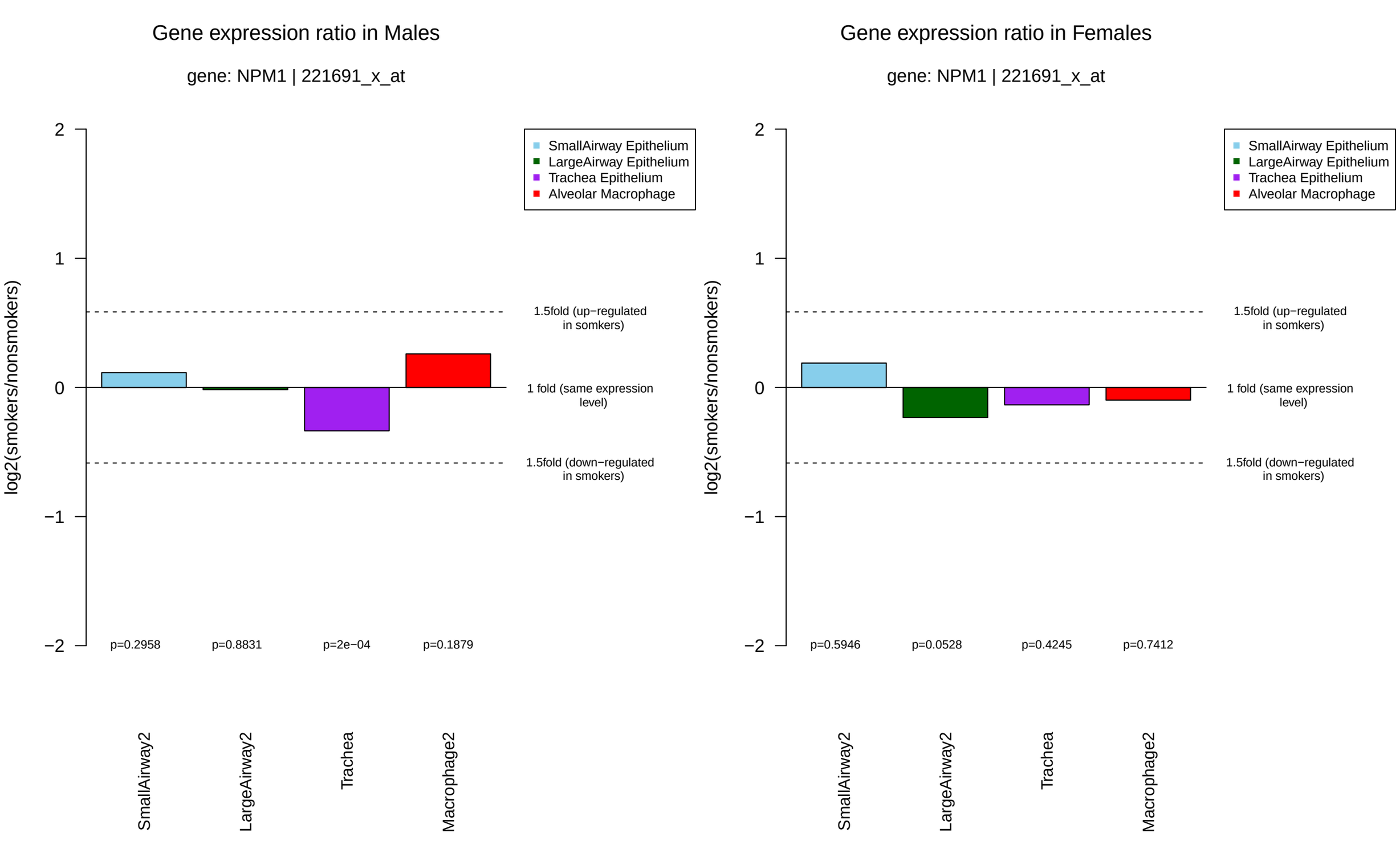
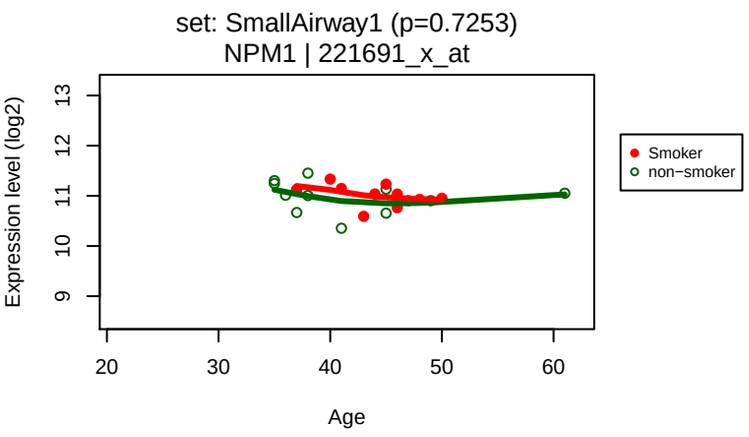
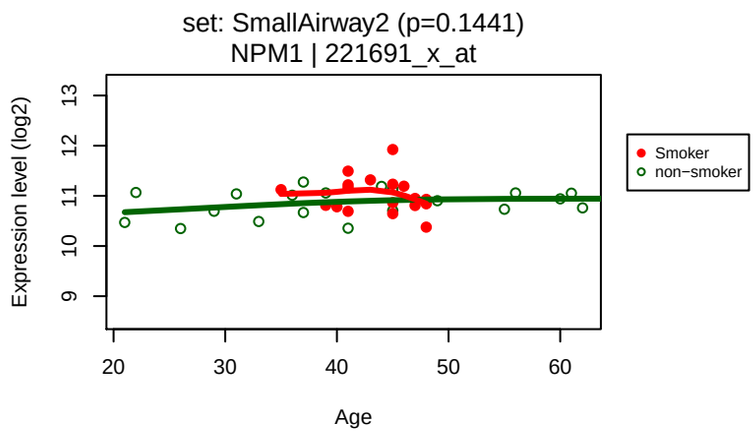
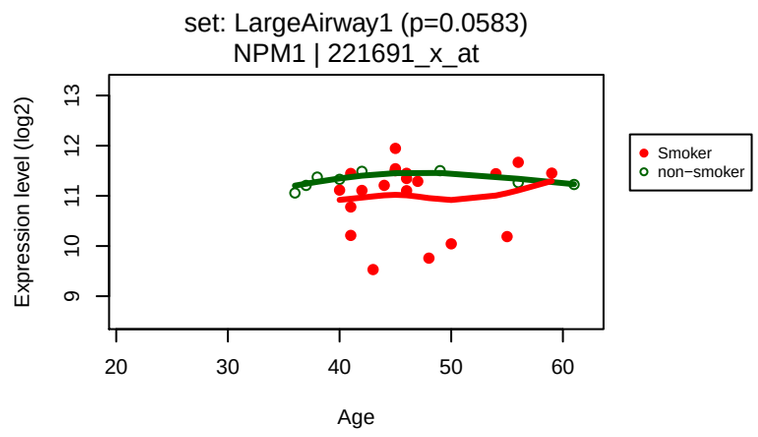

Gene expression of smokers and nonsmokers
gene: NPM1 | 221691_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7253
p=0.1441
p=0.8025
p=0.0583
p=0.3259
p=5e−04
p=0.0485
p=0.3983
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

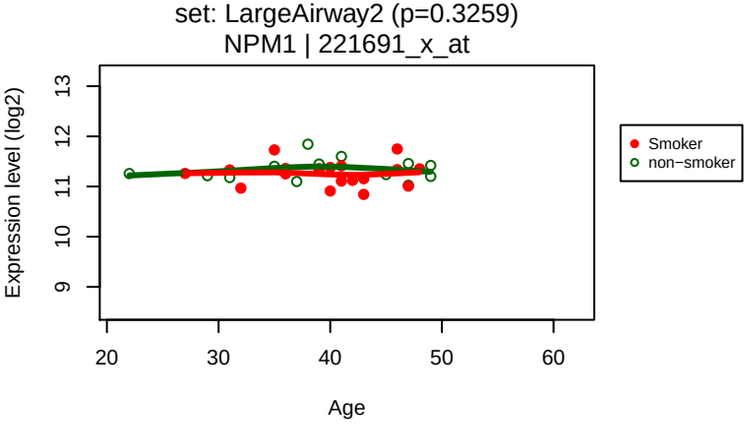

Gene expression ratio (smokers/nonsmokers)
gene: NPM1 | 221691_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7253
p=0.1441
p=0.8025
p=0.0583
p=0.3259
p=5e−04
p=0.0485
p=0.3983
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

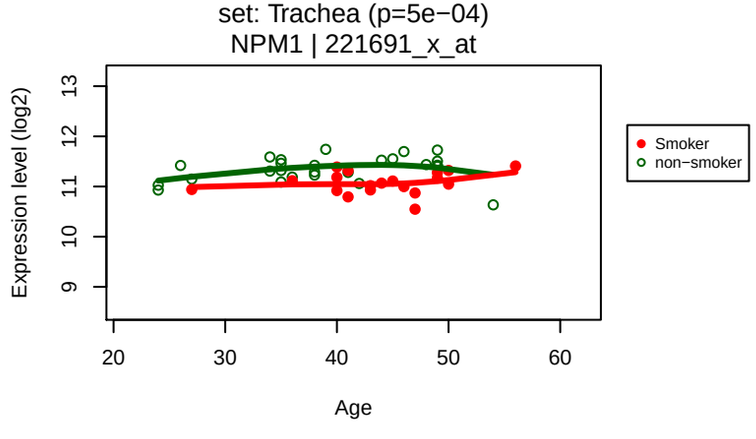

Gene expression ratio in Males
gene: NPM1 | 221691_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2958
p=0.8831
p=2e−04
p=0.1879
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPM1 | 221691_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5946
p=0.0528
p=0.4245
p=0.7412
SmallAirway2
LargeAirway2
Trachea
Macrophage2

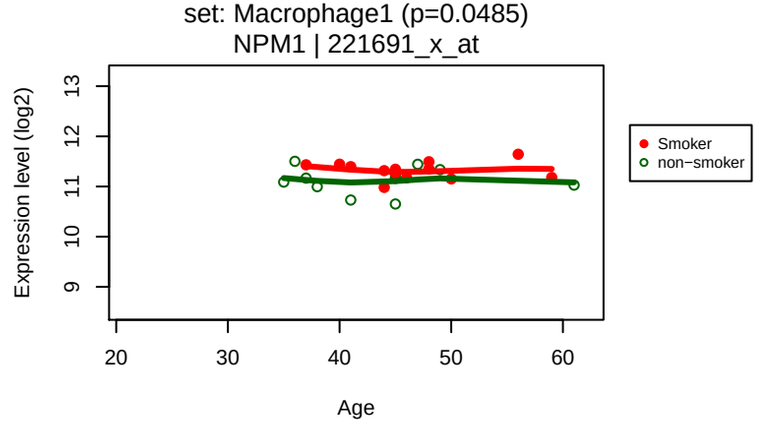

set: SmallAirway1 (p=0.7253)
NPM1 | 221691_x_at
Expression level (log2)
Age
Smoker
non−smoker

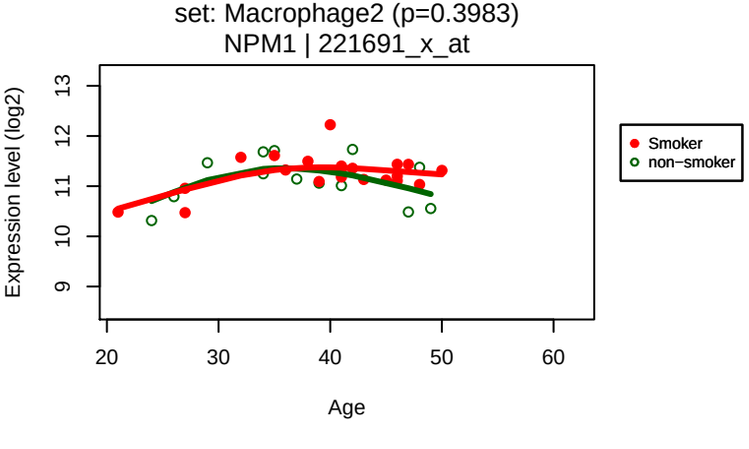

set: SmallAirway2 (p=0.1441)
NPM1 | 221691_x_at
Expression level (log2)
Age
Smoker
non−smoker

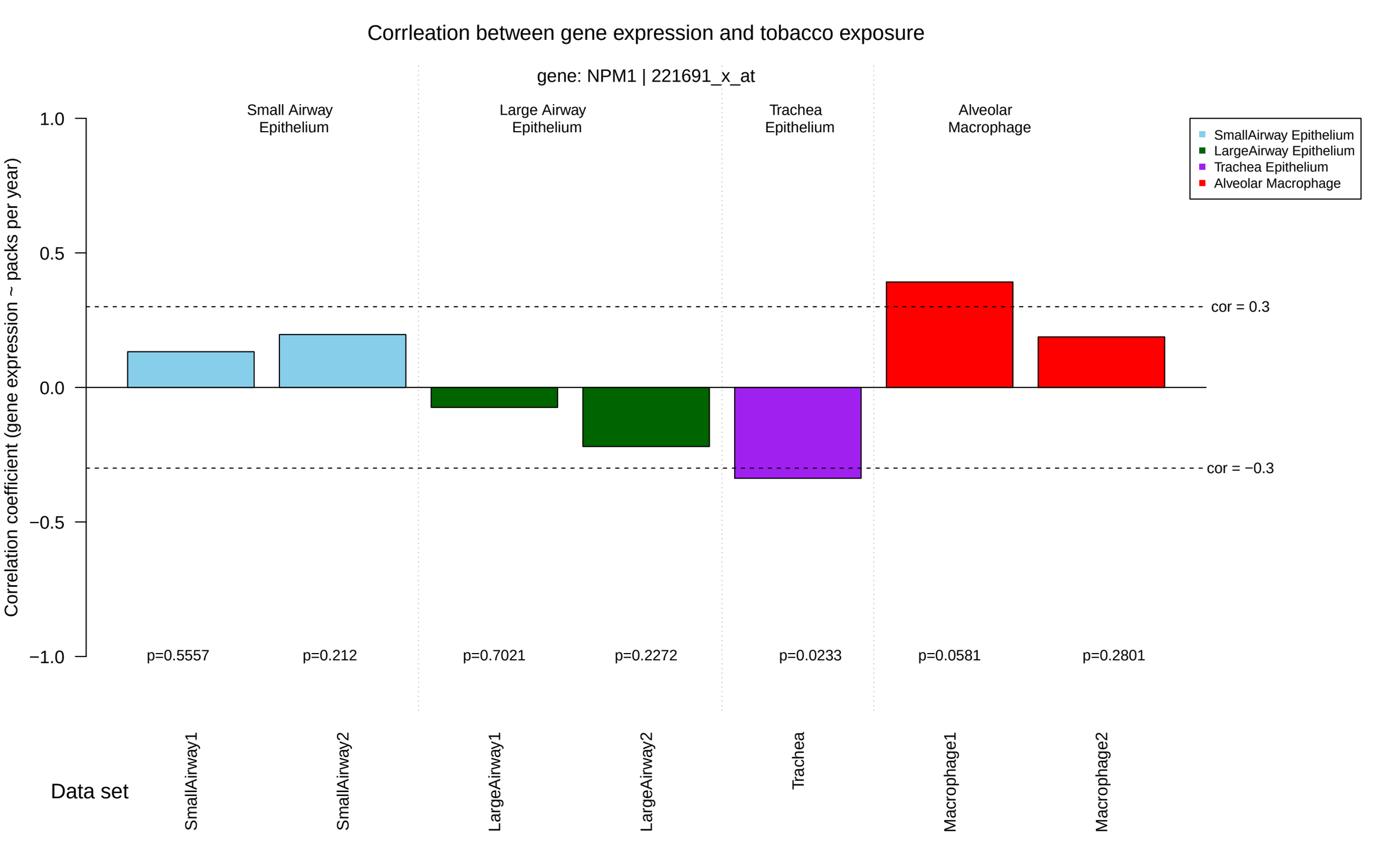

set: LargeAirway1 (p=0.0583)
NPM1 | 221691_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3259)
NPM1 | 221691_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
NPM1 | 221691_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0485)
NPM1 | 221691_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3983)
NPM1 | 221691_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPM1 | 221691_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5557
p=0.212
p=0.7021
p=0.2272
p=0.0233
p=0.0581
p=0.2801
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2